

Don't go to lecture on Friday, because there is no lecture due to inauguration.

No other announcements this week

1: $\iiint_E (xy + z^2) dV$, $E = [0, 2] \times [0, 1] \times [0, 3]$.

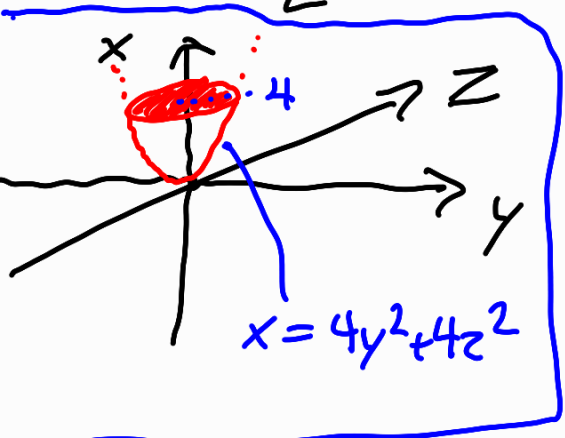
Substitute $0 \leq \overset{2}{x}, \overset{1}{y}, \overset{3}{z}$ and make sure dx, dy, dz in right order. So

$$\int_0^2 \int_0^1 \int_0^3 (xy + z^2) dz dy dx = \int_0^2 \int_0^1 \left(zxy + \frac{z^3}{3} \right) \bigg|_{z=0}^{z=3} dy dx$$

$$= \int_0^2 \int_0^1 (3xy + 9) dy dx = \int_0^2 (1.5x + 9) dx =$$

$$0.75x^2 + 9x \bigg|_0^2 = 3 + 18 = 21.$$

3: $\iiint_E x dV$, E bounded by $x=4$, $4y^2 + 4z^2$



Step 1: Graph region and find bounds from graph
So E is the region below the plane but above the

paraboloid, so $4y^2 + 4z^2 \leq x \leq 4$.

Now notice $4y^2 + 4z^2 \leq 4 \Rightarrow$ $y^2 + z^2 \leq 1$,
so (y, z) must lie inside the unit disk.

Step 2: Convert using cylindrical
 y & z are squared, so let $y = r \cos \theta$, $z = r \sin \theta$,
 $x = x$. Then $1 \geq y^2 + z^2 = r^2 \Rightarrow 1 \geq r$,
so $0 \leq r \leq 1$. Also, $4 \geq x \geq 4(y^2 + z^2) = 4r^2$.
Finally, notice E rotationally symmetric,
so no θ restrictions, so $0 \leq \theta \leq 2\pi$

Step 3: Substitute bounds & evaluate.

$$\begin{aligned} x dV &= x r dr d\theta dx, \text{ so new integral} \\ \int_0^{2\pi} \int_0^1 \int_{4r^2}^4 x r dx dr d\theta &= 2\pi \int_0^1 r \left. \frac{x^2}{2} \right|_{x=4r^2}^{x=4} dr \\ &= \pi \int_0^1 r (16 - 16r^2) dr = \pi (8r^2 - 4r^4) \Big|_0^1 \\ &= \pi (8 - 4) = \boxed{4\pi}. \end{aligned}$$

12: Find volume within $x^2 + y^2 + z^2 = 4$,
above xy -plane, below $z = \sqrt{x^2 + y^2}$.

Step 1: Find initial bounds from prepositions
within sphere: $x^2 + y^2 + z^2 \leq 4$.

Above $z=0$ plane: "above" $\Rightarrow$ $z \geq 0$

Below cone: "below" $\Rightarrow$ $z \leq \sqrt{x^2 + y^2}$

This is one case where you can avoid
graphing. May not always be possible,
especially if only equations are given
with only clues being "bounded by..."

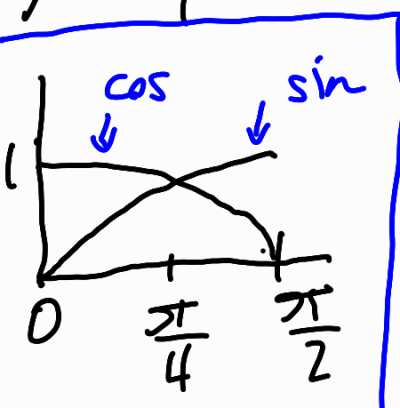
Step 2: Convert to spherical

$$x = \rho \sin \varphi \cos \theta, y = \rho \sin \varphi \sin \theta, z = \rho \cos \varphi$$

$$4 \geq x^2 + y^2 + z^2 = \rho^2 \Rightarrow \underline{0 \leq \rho \leq 2}.$$

$$0 \leq z = \rho \cos \varphi \Rightarrow \cos \varphi \geq 0 \Rightarrow 0 \leq \varphi \leq \frac{\pi}{2}$$

$$\rho \cos \varphi = z \leq \sqrt{x^2 + y^2} = \rho \sin \varphi \Rightarrow \cos \varphi \leq \sin \varphi,$$



which happens on $[0, \frac{\pi}{2}]$

when $\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{2}$ as seen

by graphing.

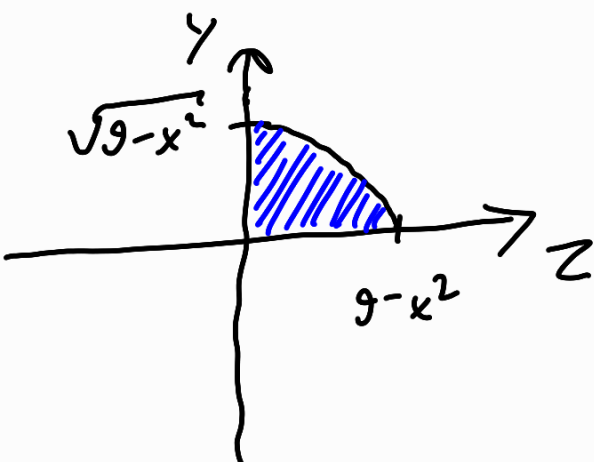
No constraint on θ , so $0 \leq \theta \leq 2\pi$.

Step 3: Substitute bounds & evaluate.

$$\begin{aligned} V &= \iiint_E dV = \int_0^{2\pi} \int_0^2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \rho^2 \sin\varphi \, d\varphi \, d\rho \, d\theta = \\ &= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^2 \rho^2 \, d\rho \right) \left(\int_{\pi/4}^{\pi/2} \sin\varphi \, d\varphi \right) = \\ &= 2\pi \cdot \frac{3^3}{3} \cdot -\cos\varphi \Big|_{\pi/4}^{\pi/2} = \frac{18\pi}{\sqrt{2}} = \boxed{9\pi\sqrt{2}} \end{aligned}$$

10: Find $\int_{-3}^3 \int_0^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} \sqrt{x^2+y^2} \, dz \, dy \, dx$

Step 1: Graph region to get new bounds



$$0 \leq y \leq \sqrt{9-x^2},$$

$$0 \leq z \leq 9-x^2-y^2 \Rightarrow$$

$$x = r \cos\theta, \, y = r \sin\theta, \text{ then } \underline{0 \leq z \leq 9-r^2},$$

$$y \leq \sqrt{9-x^2} \Rightarrow 9 \geq y^2 + x^2 = r^2 \Rightarrow r \leq 3,$$

$$\text{so } \underline{0 \leq r \leq 3}.$$

Also, $|x| \leq 3 \Rightarrow r|\cos\theta| \leq 3$. But notice that $|\cos\theta| \leq 1$ always and $r \leq 3$ from previous bound, so $r|\cos\theta| \leq 3$ always & there are no additional constraints.

Also notice $0 \leq y = r\sin\theta \Rightarrow \sin\theta \geq 0 \Rightarrow 0 \leq \theta \leq \pi$ but no other θ constraint, so $0 \leq \theta \leq \pi$.

Step 2: Plug in bounds & evaluate

$$\sqrt{x^2+y^2} dz dy dx = r dV = r^2 dr d\theta dz,$$

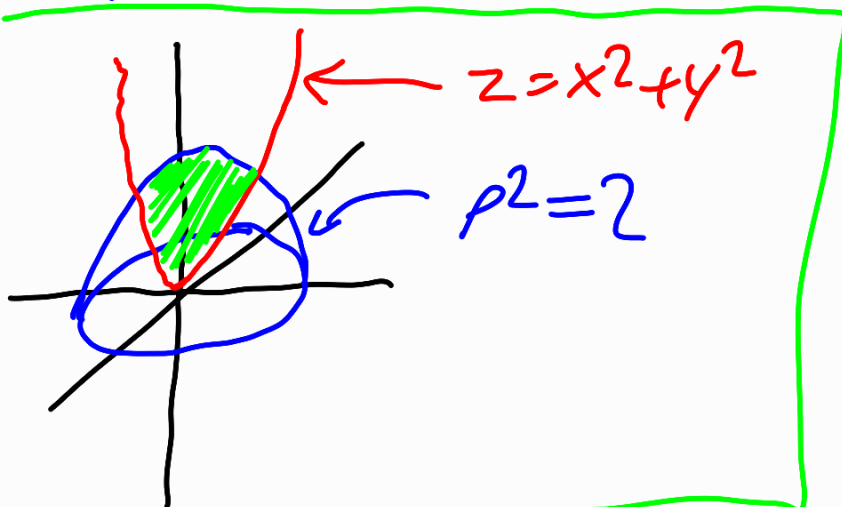
so new integral is

$$\int_0^\pi \int_0^3 \int_0^{9-r^2} r^2 dz dr d\theta = \pi \int_0^3 r^2(9-r^2) dr$$

$$= \pi \left(3r^3 - \frac{r^5}{5} \right) \Big|_0^3 = \pi \left(81 - \frac{243}{5} \right) =$$

$$\pi \cdot \frac{162}{5} = \frac{162\pi}{5}.$$

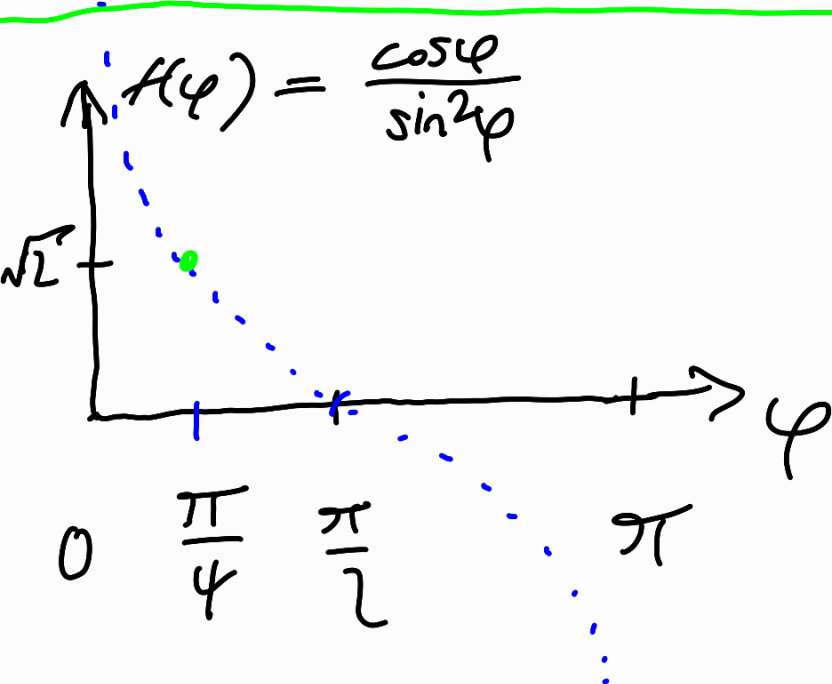
8: Volume between $z = x^2 + y^2$ & $x^2 + y^2 + z^2 = 2$
 Step 1: Since there is a sphere, use spherical



$$\begin{aligned}x &= \rho \sin \varphi \cos \theta \\y &= \rho \sin \varphi \sin \theta \\z &= \rho \cos \varphi\end{aligned}$$

From the graph of the region, we know

$\rho^2 \leq 2$ and $z \geq x^2 + y^2$. First bound $\Rightarrow$
 $0 \leq \rho \leq \sqrt{2}$. 2nd one $\Rightarrow \rho^2 \sin^2 \varphi \leq \rho \cos \varphi$
 $\Rightarrow \rho \sin^2 \varphi \leq \cos \varphi \Rightarrow \rho \leq \frac{\cos \varphi}{\sin^2 \varphi} := f(\varphi)$



Notice that
 $f \leq 0$ on $[\frac{\pi}{2}, \pi]$,
 so $0 \leq \varphi \leq \frac{\pi}{2}$
 only since we
 need $\rho \geq 0$. Also

notice $f(\frac{\pi}{4}) = \sqrt{2}$, so
 $0 \leq \varphi \leq \frac{\pi}{4}$, otherwise

$\rho \leq \sqrt{2}$ for
 $\rho \leq f(\varphi)$. So

$$0 \leq \rho \leq \left\{ \begin{array}{l} \sqrt{2}, 0 \leq \varphi \leq \frac{\pi}{4} \\ \frac{\cos \varphi}{\sin^2 \varphi}, \frac{\pi}{4} \leq \varphi \leq \frac{\pi}{2} \end{array} \right\} := g(\varphi). \text{ Also}$$

no restriction on θ , so $0 \leq \theta \leq 2\pi$.

Step 2: Plug in the bounds & evaluate 2 separate integrals for piecewise g .

$$V = \iiint_{\text{region}} dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{g(\varphi)} \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$$

$$= 2\pi \int_0^{\pi/2} \sin \varphi \frac{g(\varphi)^3}{3} \, d\varphi = \frac{2\pi}{3} \underbrace{\int_0^{\pi/2} g^3(\varphi) \sin \varphi \, d\varphi}_I$$

$$I = \int_0^{\pi/4} (\sin \varphi)^2 \sqrt{2} \, d\varphi + \int_{\pi/4}^{\pi/2} \frac{\cos^3 \varphi}{\sin^5 \varphi} \, d\varphi := A + B$$

$$A = -2\sqrt{2} \cos \varphi \Big|_0^{\pi/4} = 2\sqrt{2} \left(1 - \frac{1}{\sqrt{2}}\right) = 2\sqrt{2} - 2$$

$$B = \int_{\pi/4}^{\pi/2} \frac{\cos^2 \varphi \cos \varphi}{\sin^5 \varphi} \, d\varphi = \int_{1/\sqrt{2}}^1 \frac{1-u^2}{u^5} \, du$$

using $u = \sin \varphi \Rightarrow du = \cos \varphi \, d\varphi$ and

$$\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{2} \Rightarrow \frac{1}{\sqrt{2}} \leq \sin \varphi \leq 1. \quad \text{So}$$

$$B = \int_{1/\sqrt{2}}^1 (u^{-5} - u^{-3}) du = \left. -\frac{u^{-4}}{4} + \frac{u^{-2}}{2} \right|_{1/\sqrt{2}}^1 =$$

$$\left(-\frac{1}{4} + \frac{1}{2}\right) - \left(-1 + 1\right) = \frac{1}{4}, \quad \text{so}$$

$$I = A + B = 2\sqrt{2} - 2 + \frac{1}{4} = 2\sqrt{2} - \frac{7}{4},$$

$$\text{so} \quad V = \frac{2\pi}{3} I = \frac{\pi}{6} (8\sqrt{2} - 7).$$